

2(a)(i)	1. $W = mas$	B1
	2. $s = (v^2 - u^2) / 2a$	B1
2(a)(ii)	W/work equals energy transferred/ <u>gain or change in</u> kinetic energy	B1
	$W (= mas) = ma(v^2 - u^2) / 2a$	B1
	leading to $W = m(v^2 - u^2) / 2$ (so $KE = \frac{1}{2}mv^2$)	
2(b)(i)	1. solid curved line drawn from X to Y along path of ball and labelled D	B1
	2. solid straight line drawn from X to Y and labelled S	B1
2(b)(ii)	$(\Delta)E = mg(\Delta)h$	C1
	$4.5 = (0.040 \times 9.81 \times h) + (\frac{1}{2} \times 0.040 \times 9.5^2)$	C1
	$h = 6.9 \text{ m}$	A1
2(b)(iii)	line with a negative gradient starting from a non-zero value of kinetic energy when the vertical height is zero	M1
	<u>straight</u> line ends at a non-zero value of kinetic energy when the vertical height is h	A1